Sahaj Bhadja

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Summary

Software engineer with 2+ years building **production AI systems**, **multi-model LLM orchestration**, and **full-stack applications** on cloud platforms. Experienced with RAG pipelines, conversational AI workflows, and deploying scalable systems that serve enterprise users.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4.0*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4.0*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*

- Built **production-grade AI applications orchestrating multiple LLMs** (GPT-4o, Claude, vision models) with switchable model architectures and intelligent routing, handling 10K+ daily requests with optimized tool selection.
- Developed **agentic workflows** for automated content analysis, implementing reasoning loops that reduced feedback turnaround by **60%** while maintaining conversation context across multi-turn interactions.

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*

- Built **full-stack applications on cloud infrastructure** (Azure) serving 500+ users, implementing APIs that connect to enterprise databases and reducing latency from **1.2s to 350ms** through optimization.
- Developed **data pipelines and observability frameworks** processing 5K+ monthly jobs with real-time monitoring, error tracing, and alerting to identify and resolve failures across distributed services.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 – May 2025*

- Built **AI/ML experimentation pipelines** using PyTorch, implementing model evaluation frameworks that tracked reasoning performance and optimized inference across 100K+ records with reproducible results.
- Collaborated with research teams to **optimize model architectures**, understanding context windows and retrieval strategies to improve planning accuracy by 18% through systematic experimentation.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*

- Engineered **cloud-native data pipelines on AWS** (S3, Glue, Lambda) with infrastructure-as-code patterns, implementing automated deployment workflows that improved pipeline efficiency by **68%**.
- Built **REST APIs connecting to enterprise systems** and maintained CI/CD pipelines with automated testing, participating in code reviews following Google-style engineering best practices in Agile sprints.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*

- Developed **full-stack applications integrating with enterprise IT infrastructure**, building APIs for 8 microservices handling 2M+ records with connections to legacy databases and security perimeters.
- Implemented **real-time data synchronization** via WebSockets and event-driven patterns, enabling live updates across distributed systems with tracing for debugging cross-service interactions.

Skills

AI/Agentic Systems: LLM orchestration (GPT-4, Claude, Gemini), RAG pipelines, multi-model routing, conversational AI
Cloud Platforms: AWS (S3, Glue, Lambda, ECS), Azure, GCP basics, Docker, infrastructure-as-code patterns
Languages & APIs: Python, TypeScript, REST APIs, FastAPI, Flask, event-driven architectures
Data & Observability: PostgreSQL, MongoDB, Redis, FAISS, Pinecone, monitoring, distributed tracing
Practices: Git, CI/CD, Agile/Scrum, code reviews, microservices, enterprise system integration

Projects

Big-Brain Live | *LLaMA-2, FAISS, Pinecone, PostgreSQL, GCP*

- Built a **RAG-powered conversational AI platform** with optimized chunking strategies and retrieval pipelines, achieving 95% query accuracy across 50K+ documents while preventing hallucinations through context grounding.

Linguistix Code | *Python, AWS (ECS, Fargate, CodePipeline), Docker, Terraform*

- Built **cloud-native deployment pipelines on AWS** using infrastructure-as-code patterns with ECS/Fargate, implementing automated resource provisioning and validation gates for production-grade reliability.

HMIS Code | *Python, FastAPI, React, PostgreSQL, Docker*

- Built an **enterprise-grade application** with complex multi-step workflows, connecting frontend interfaces to backend APIs and legacy database systems while maintaining security and role-based access controls.